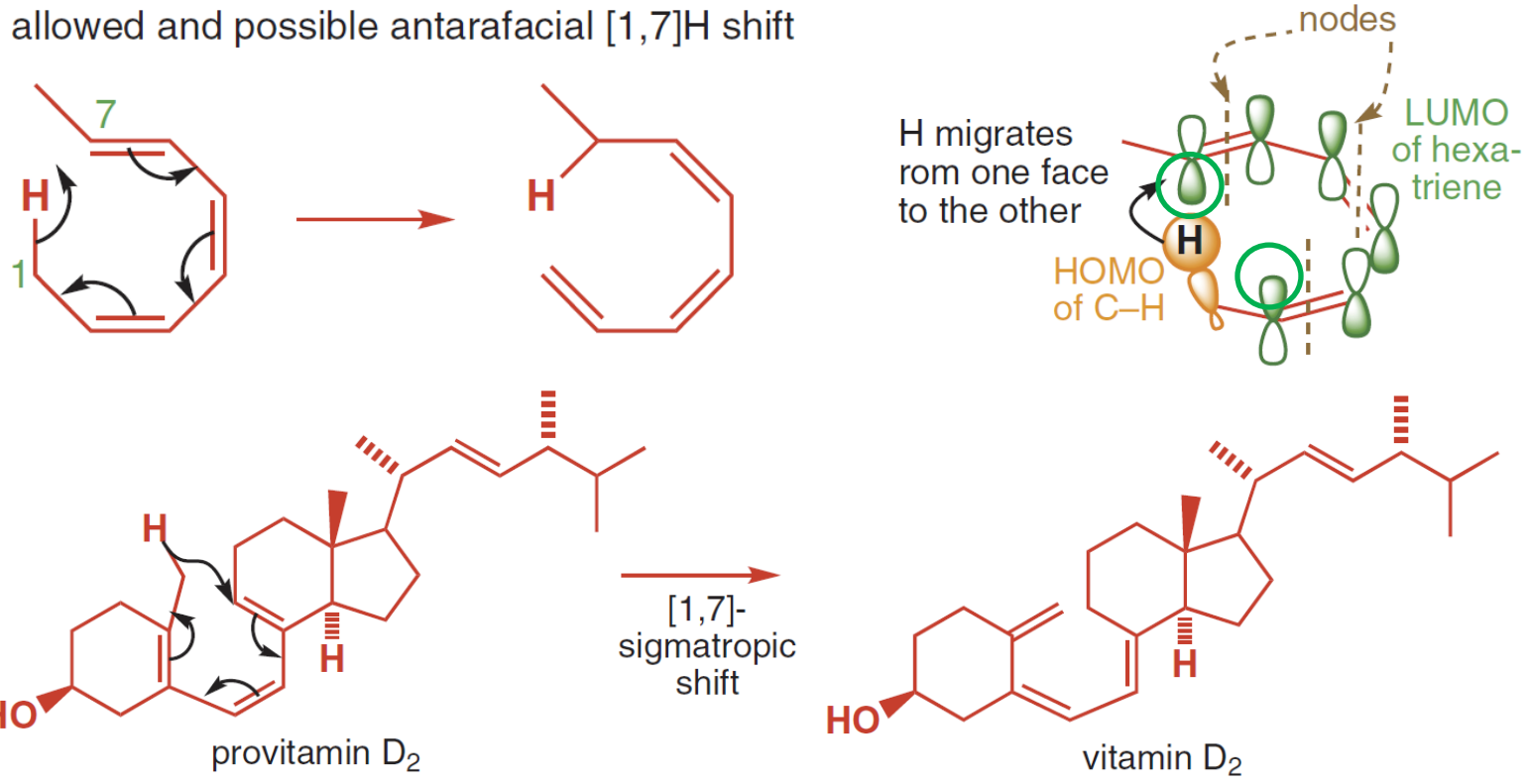
7. [1,7]-Sigmatropic hydrogen shifts (thermal: antarafacial)

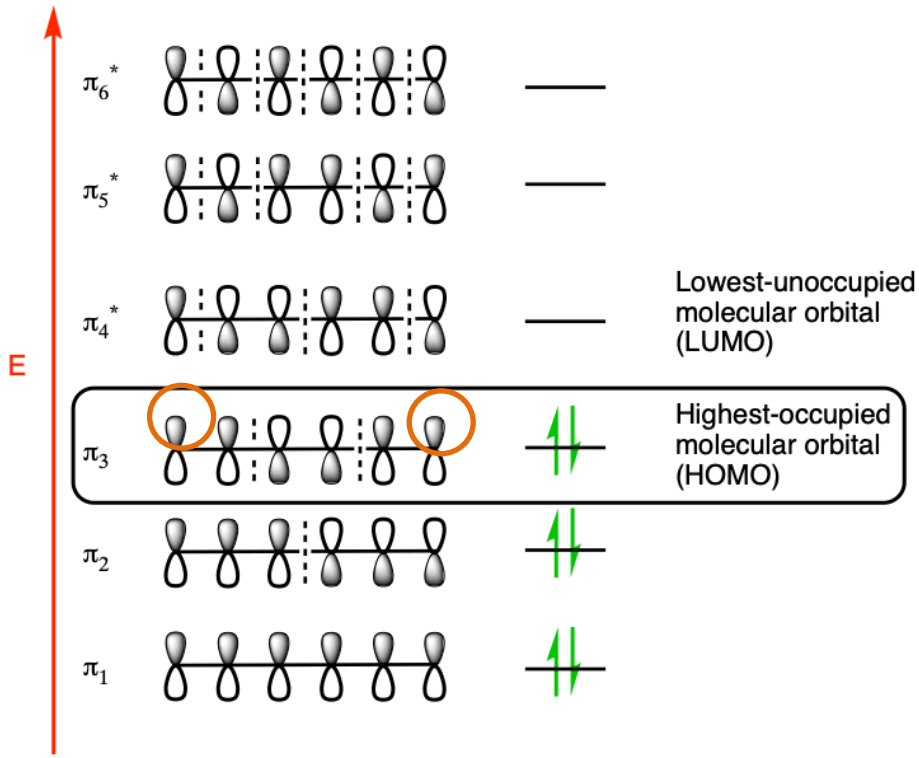


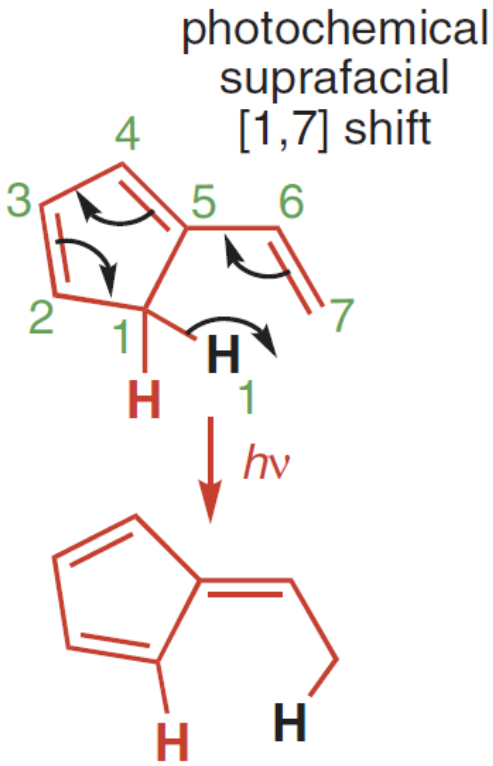
8. Photochemical [1,7] sigmatropic H shift (suprafacial)

- Photochemical and thermal reactions follow the opposite rules.

What are the molecular orbitals of excited state?

Photochemical conditions: HOMO-HOMO interactions





(thermal [1,5] H shift is also possible)

CM3121 – Advanced Organic Chemistry

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Electrocyclic reactions

The Woodward-Hoffmann rules

The rules are inversed between thermal and photochemical reactions.

- **In a thermal pericyclic reaction the total number of (4q + 2)s and (4r)a components must be odd.**
- **In a photochemical pericyclic reaction the total number of (4q + 2)s and (4r)a components must be even.**

Reaction	Approach	Components to be counted	Conditions that allow the reaction to take place
[2+2]	suprafacial-suprafacial	1+1=2	photochemical
[2+2]	suprafacial-antarafacial (ketene)	1+0=1	thermal
[1,7] H shift	suprafacial (C-H)-suprafacial (triene)	1+1=2	photochemical
[1,7] H shift	suprafacial (C-H)-antarafacial (triene)	1+0=1	thermal

Pericyclic reactions

The “simplified” rules derived from Frontier molecular orbitals and Woodward-Hoffmann rules

- Total number of participating electrons:
  - Thermal:
    - All suprafacial components: (4n+2) allowed, e.g., [4+2] cycloaddition, [3+2] dipolar cycloaddition, [6+4] cycloaddition, [1,5] H shift, [3,3] rearrangement
    - with one antrafacial component: (4n) allowed, e.g., thermal [2+2] (suprafacial-antarafacial), [1,7] H shift
  - Photochemical:
    - All suprafacial components: (4n) allowed, e.g. photochemical [2+2] cycloaddition, photochemical [1,7] H shift
    - with one antrafacial component: (4n+2) allowed

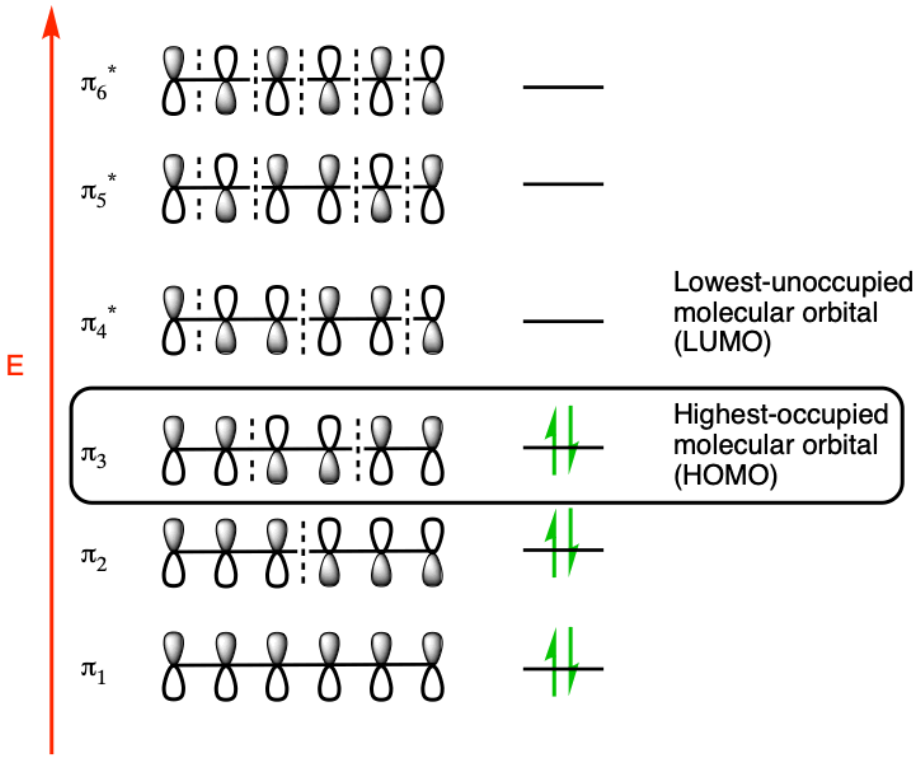
n = a whole number (0, 1, 2, 3…)

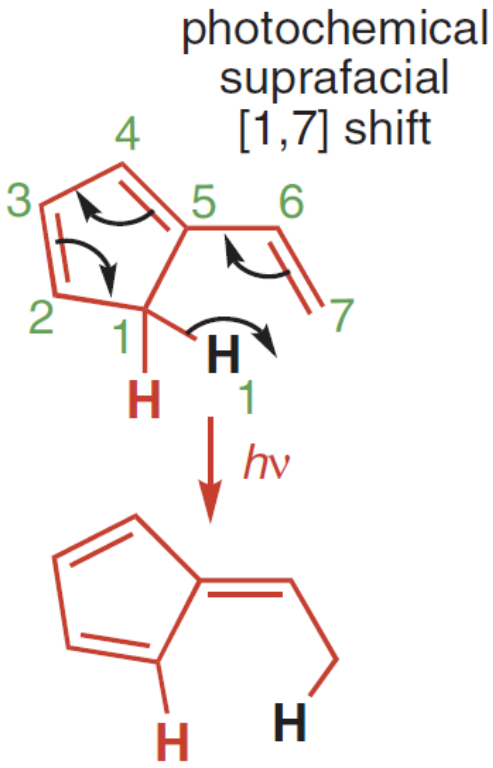
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Photochemical conditions: HOMO-HOMO interactions

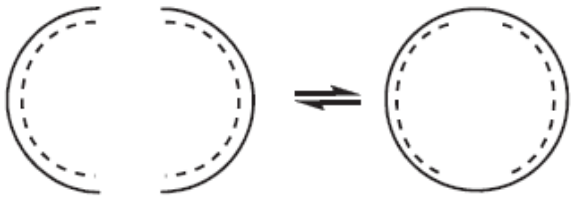




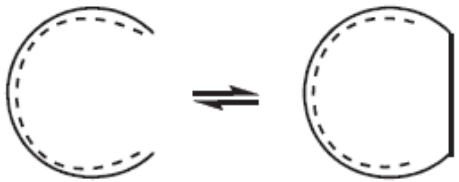
(why not photochemical [1,5] H shift?)

Pericyclic reactions (classifications)

- Cycloadditions
- Electrocyclic reactions
  - One π system
  - Allowed based on Woodward-Hoffmann rules
    - Thermal: **1** component fits the description.
    - Photochemical: **0** component fits the description.
- Sigmatropic rearrangements



(a) Cycloadditions



(b) Electrocyclic reactions



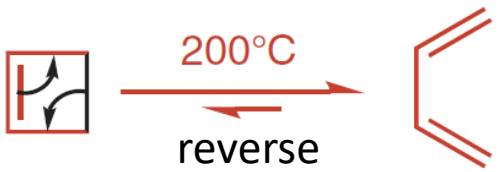
(c) Sigmatropic rearrangements

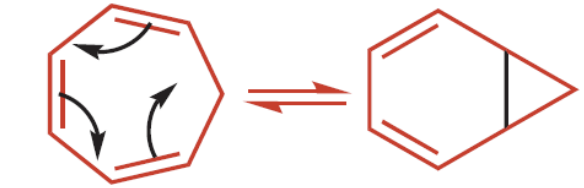
1. Description of electrocyclic reactions

An electrocyclic reaction is the formation of a new σ bond across the ends of a conjugated polyene or the reverse

- Ring formation or opening depending on the thermodynamics
  - forward reactions
  - reverse reactions (electrocyclic cleavage)
- Conjugated π system (1 component)
- Photochemical or thermal
- Disrotatory or conrotatory



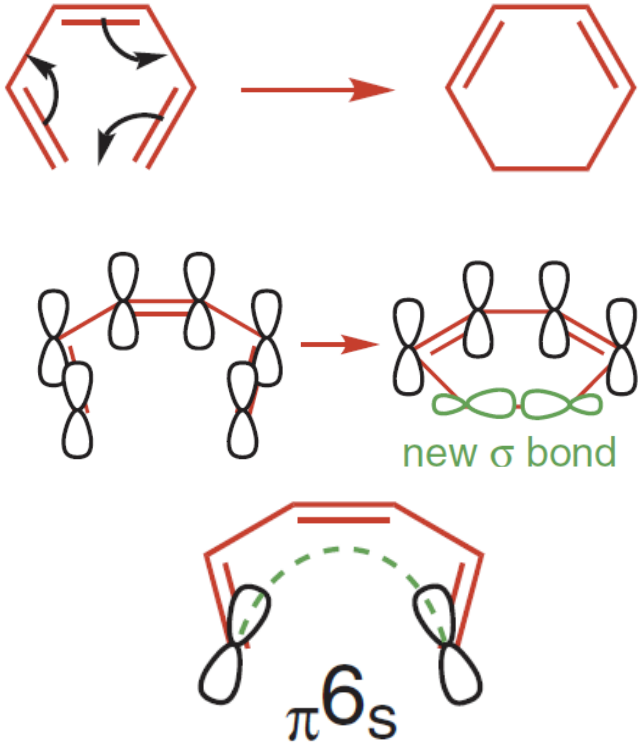




2. The Woodward–Hoffman rules

- **In a thermal pericyclic reaction the total number of (4q + 2)s and (4r)a components must be odd.**

6 π system

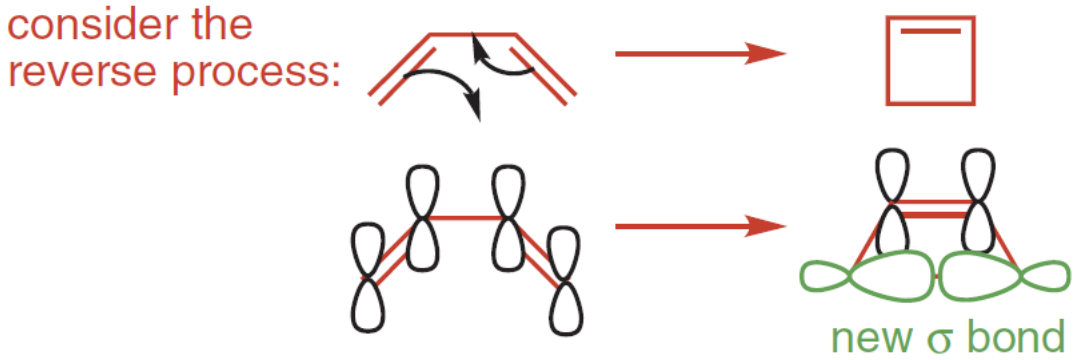


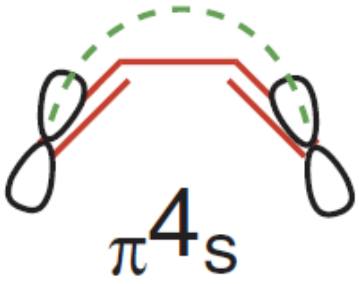
allowed reaction

The Woodward–Hoffman rules

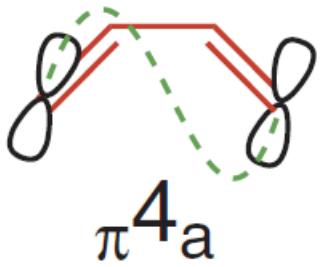


It is easier to consider the ring-closing reaction even if ring opening is favoured thermodynamically. They share the same transition state.





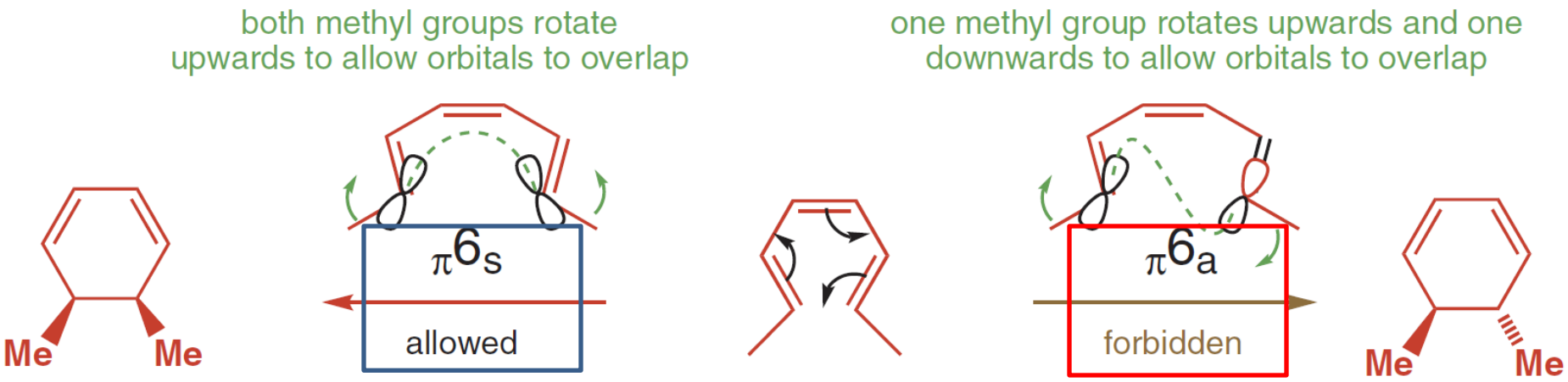
disallowed reaction



allowed reaction

3. Disrotatory Electrocyclic reactions

- Thermal electrocyclic reactions involving (4n + 2) π electrons are disrotatory



disrotatory reactions: one group rotates clockwise and one anticlockwise.

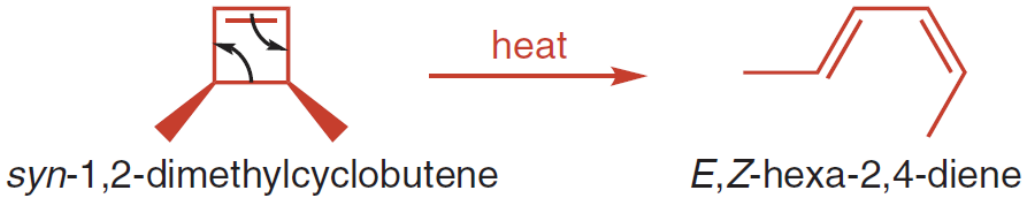
(suprafacial)

conrotatory reactions: the two groups rotate both clockwise or both anticlockwise.

(antarafacial)

Conrotatory Electrocyclic reactions

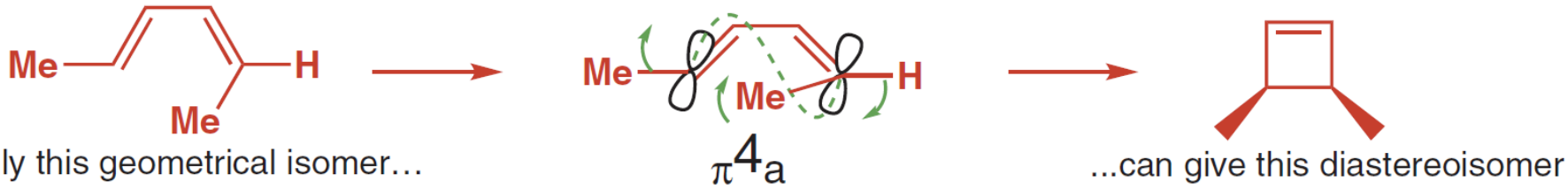
- Thermal electrocyclic reactions involving (4n) π electrons are conrotatory



syn-1,2-dimethylcyclobutene

E,*Z*-hexa-2,4-diene

both methyl groups rotate upwards
to allow orbitals to overlap



only this geometrical isomer…

π^4_a

…can give this diastereoisomer

conrotatory reactions: the two groups rotate both clockwise or both anticlockwise.

(antarafacial)

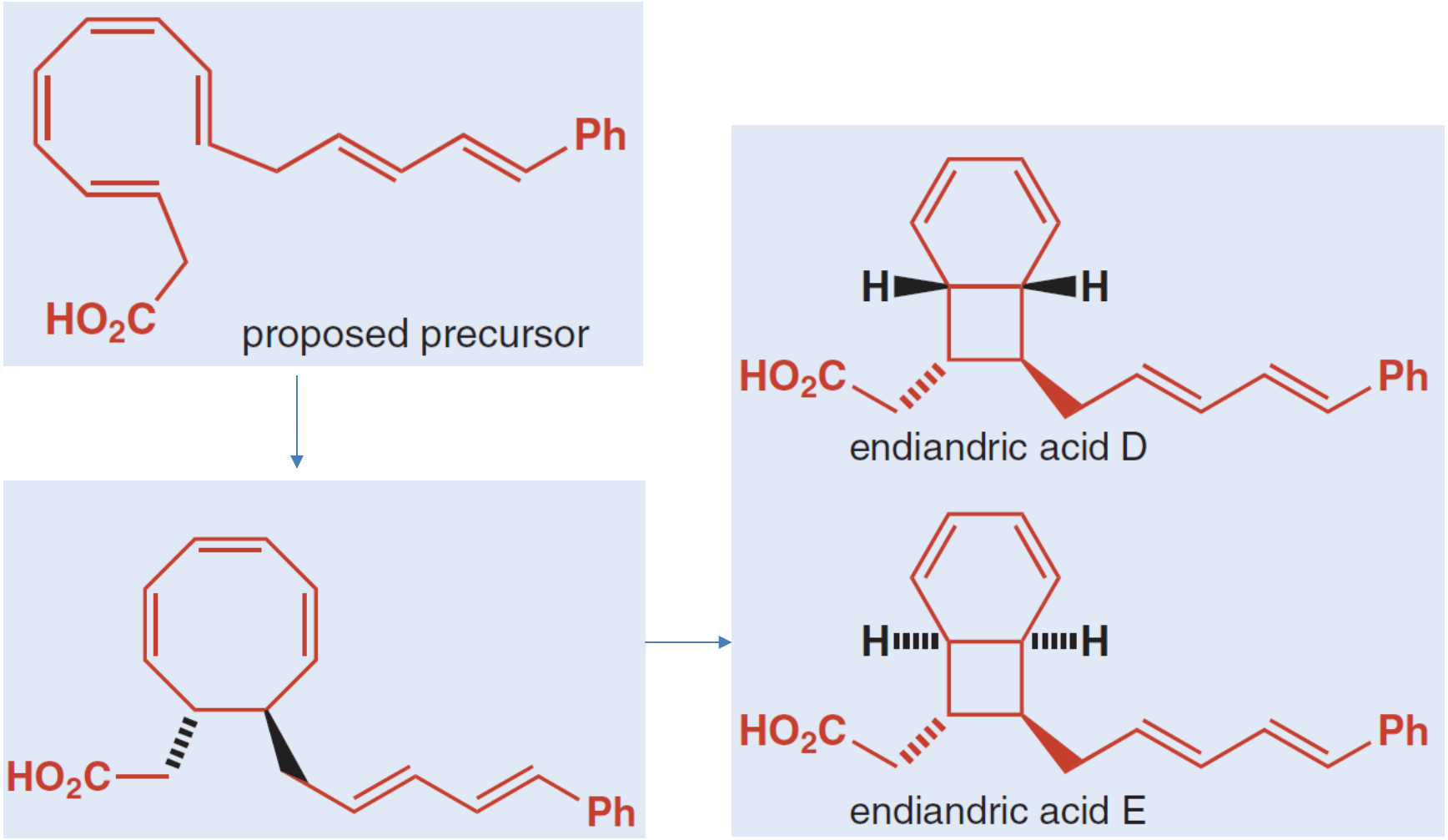
Electrocyclic reactions

Thermal conditions:

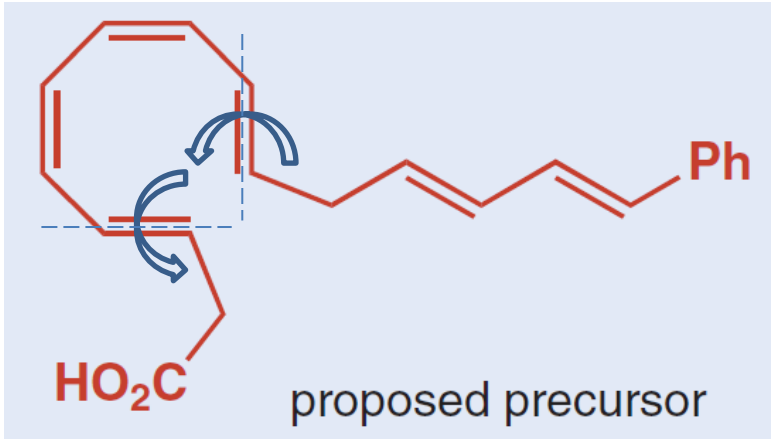
- Reactions involving (4n + 2) π electrons are disrotatory.
- Reactions involving (4n) π electrons are conrotatory.

The endiandric acids

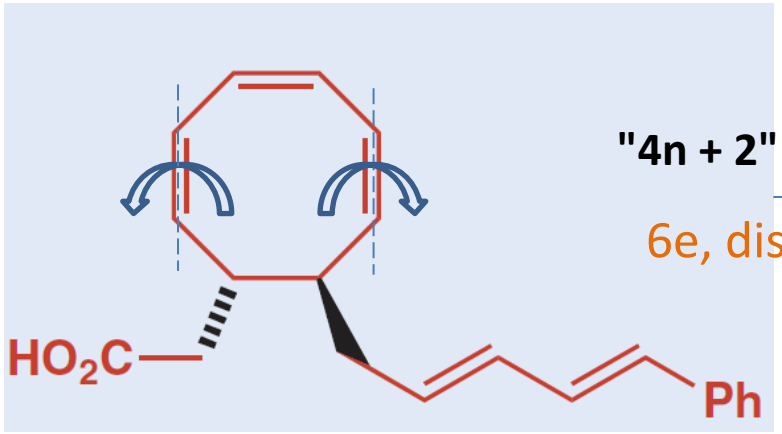
Explain the stereochemistry under thermal conditions.



The endiandric acids



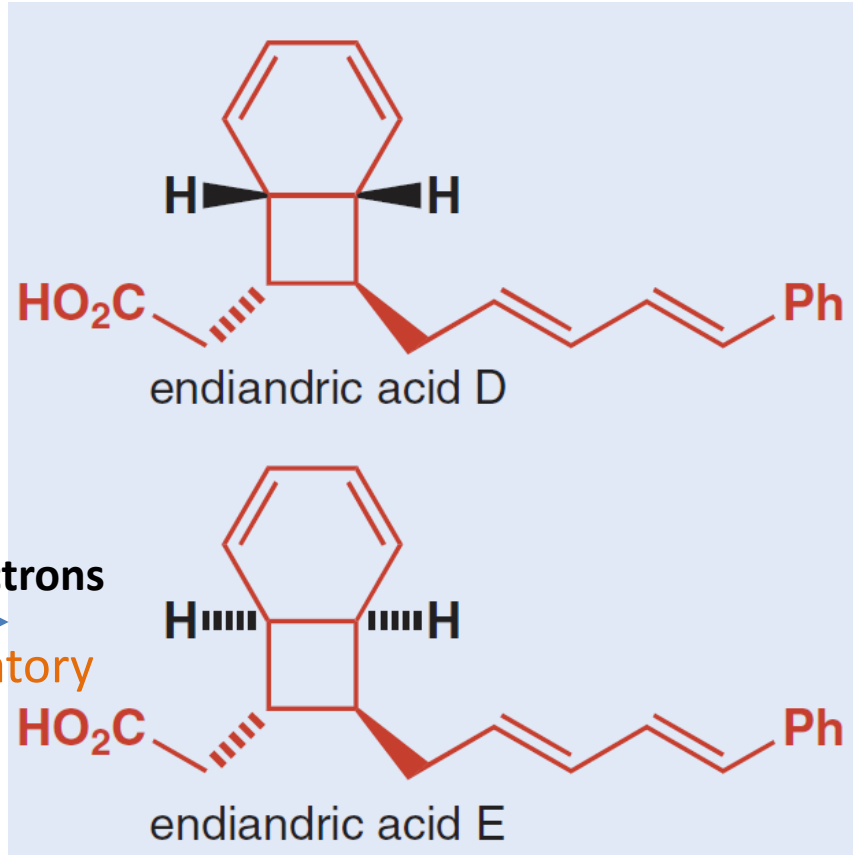
"4n" electrons | 8e, conrotatory



"4n + 2" electrons

6e, disrotatory

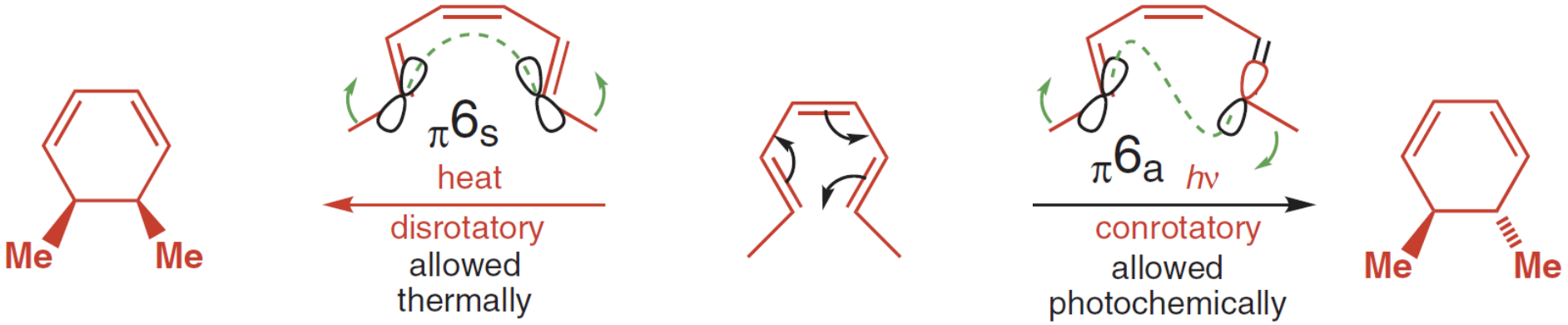
Explain the stereochemistry under thermal conditions.



4. Photochemical electrocyclic reactions

Photochemical conditions:

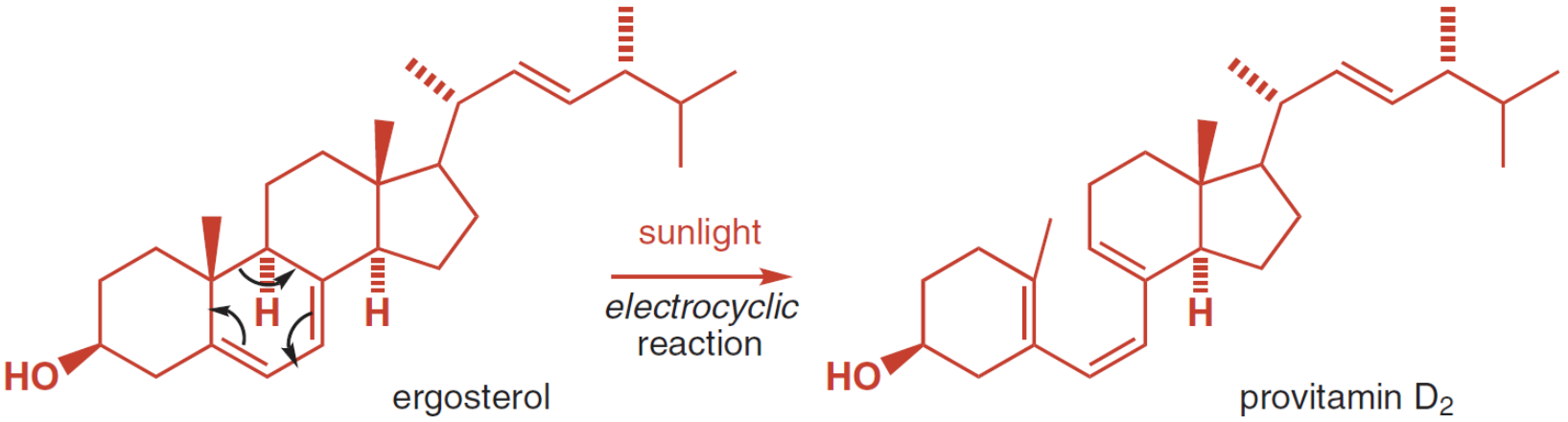
- Reactions involving (4n + 2) π electrons are conrotatory.
- Reactions involving (4n) π electrons are disrotatory.



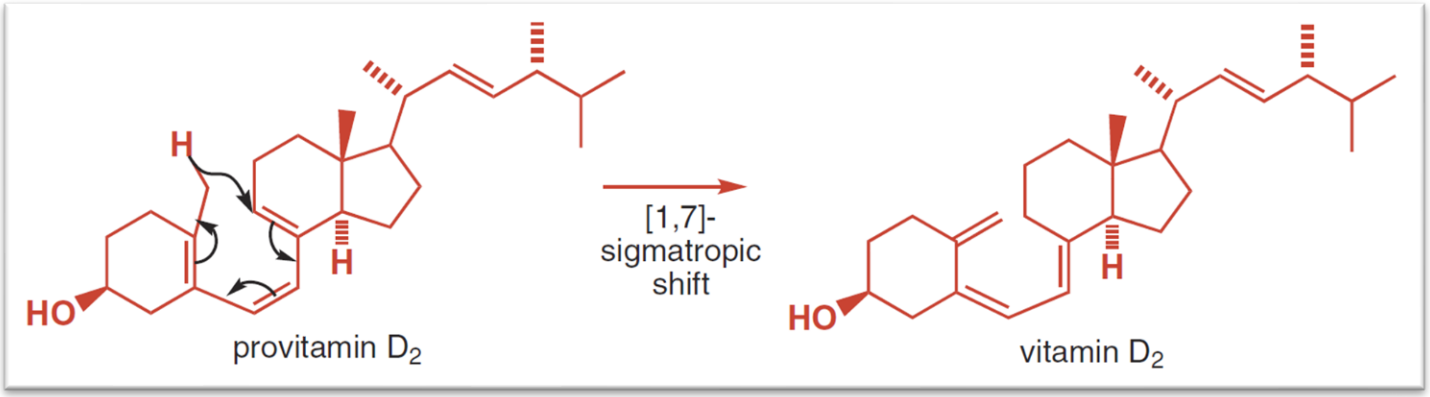
The rules are inversed between thermal and photochemical reactions.

Vitamin D

- Electrocyclic reactions are allowed both thermally and photochemically.
- The steric requirements for con- or disrotatory cyclization or ring opening may make one or both modes impossible.

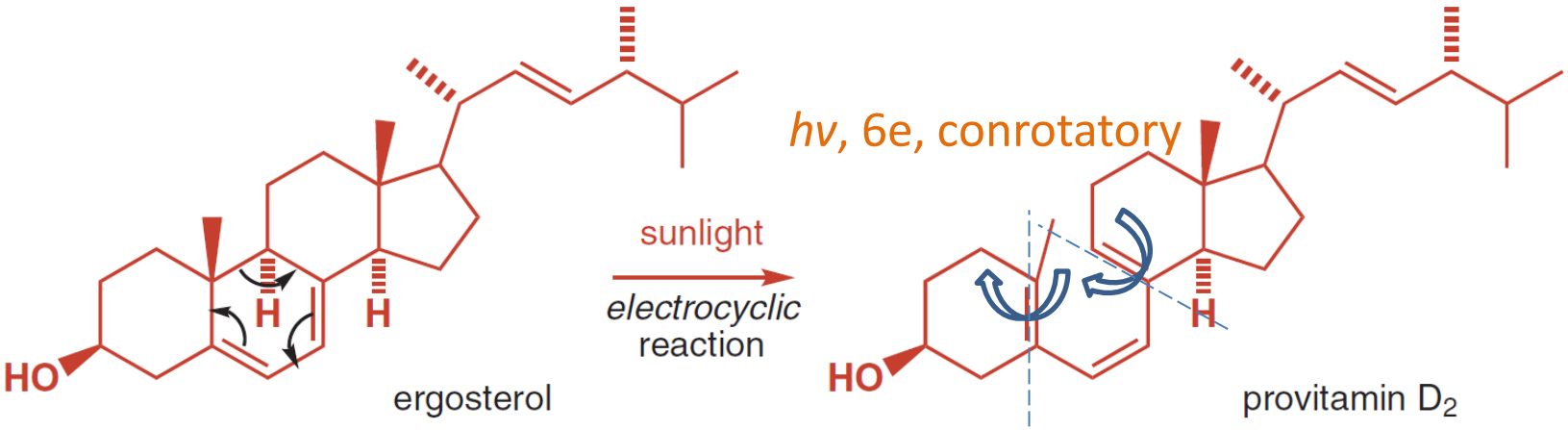


Why is sunlight required? (Hint: consider the reverse reaction)

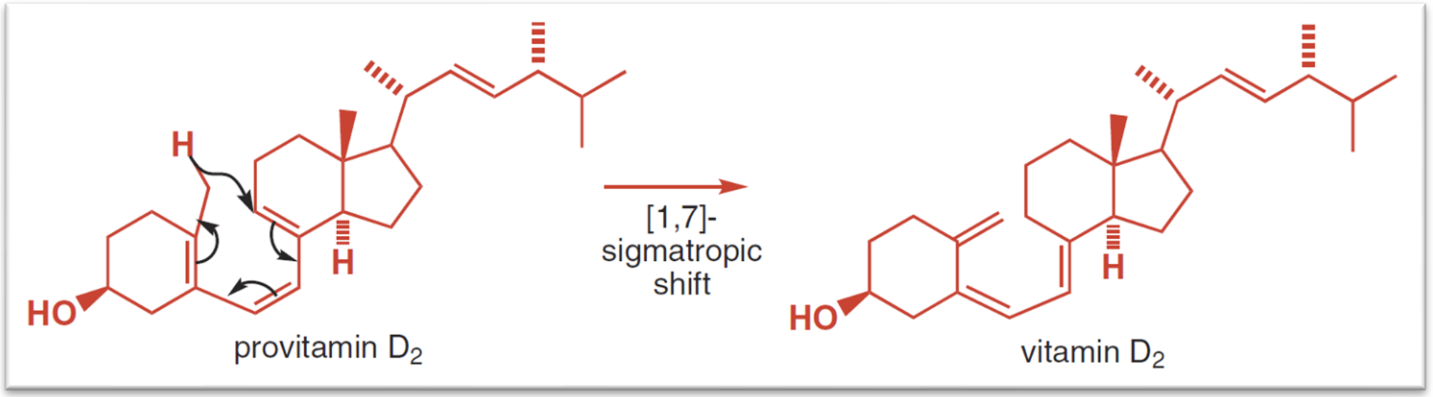


Vitamin D

- Electrocyclic reactions are allowed both thermally and photochemically.
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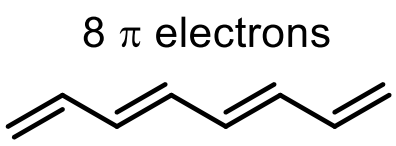


Why is sunlight required? (Hint: consider the reverse reaction)



Molecular orbitals vs. Woodward–Hoffman rules

8 π electrons



Consider the frontier molecular orbital (FMO).

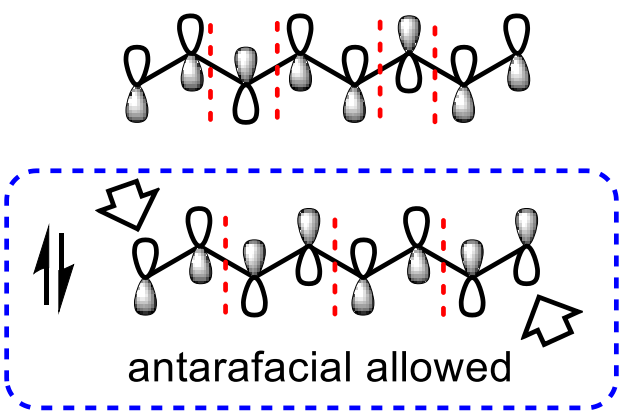
8 π thermal
Conrotatory: **antarafacial**

8 π photochemical
Disrotatory : **suprafacial**

number of nodes

4 nodes (LUMO)

3 nodes (HOMO)



8 π antarafacial: 1 qualifying component
1 is odd, so thermal reaction is allowed.

In a thermal pericyclic reaction the total number of (4q + 2)s and (4r)a components must be odd.

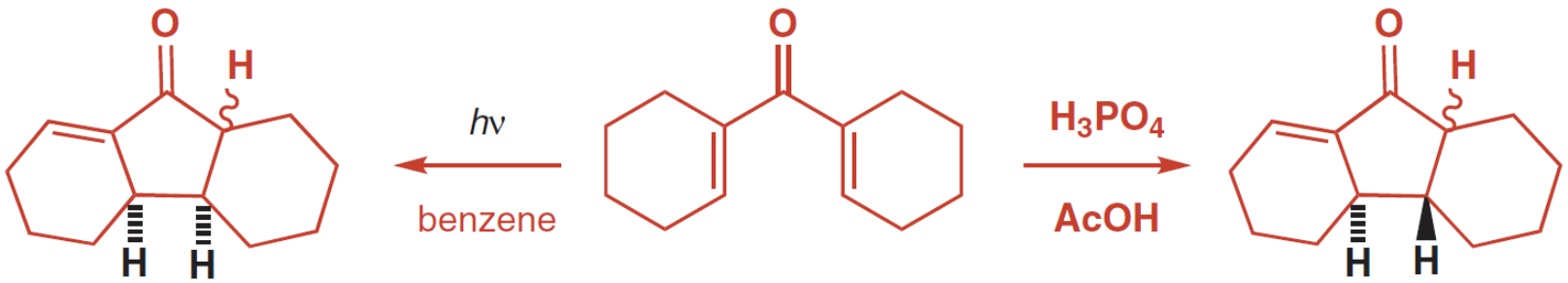
8 π suprafacial: 0 qualifying component
0 is even, so photochemical reaction is allowed.

In a photochemical pericyclic reaction the total number of (4q + 2)s and (4r)a components must be even.

5. Nazarov cyclizations

- Doubly α,β-unsaturated ketone can be converted to a cyclopentenone under acidic conditions.

Is the reaction disrotatory or conrotatory?

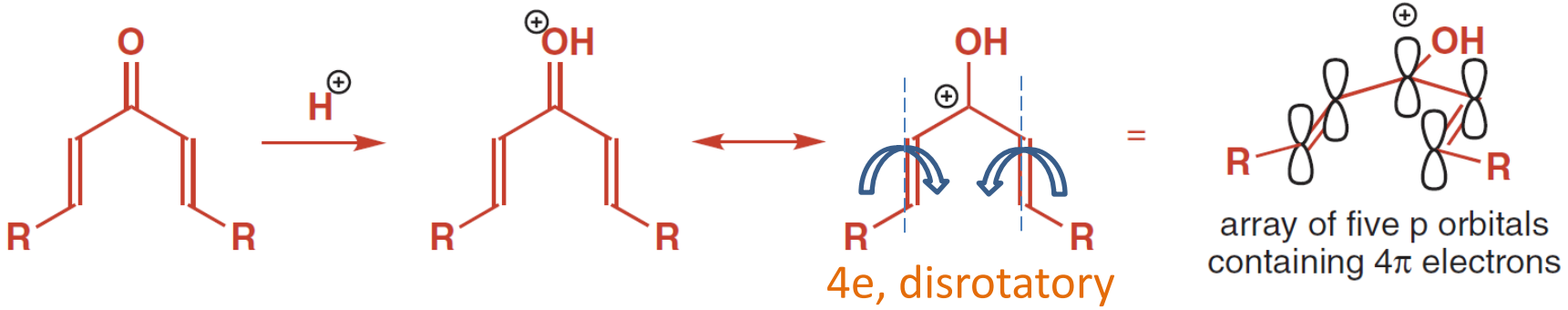


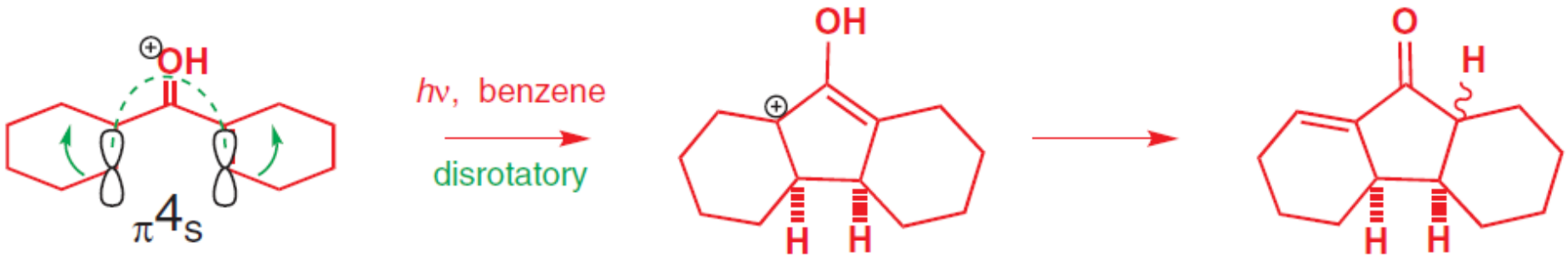
Explain the stereochemistry of 1. photochemical conditions and 2. thermal conditions

Nazarov cyclizations: stereochemistry

Is the reaction disrotatory or conrotatory?

- Photochemical conditions

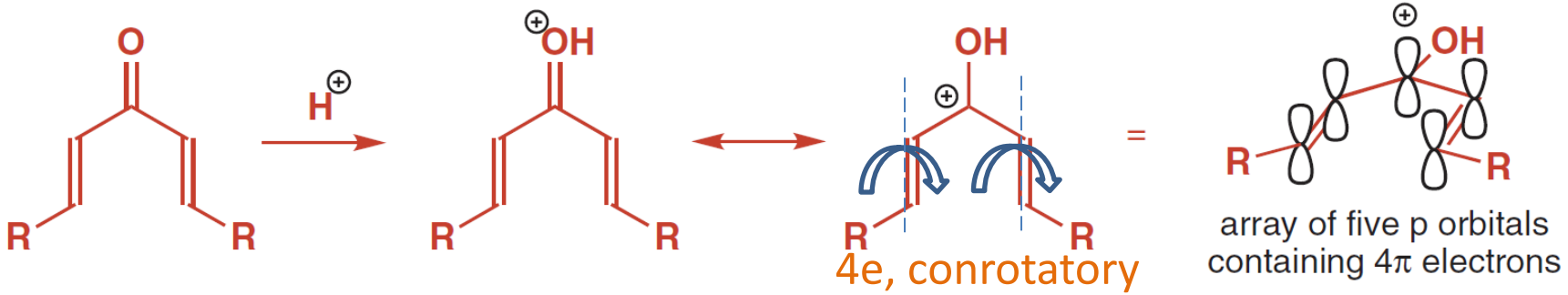


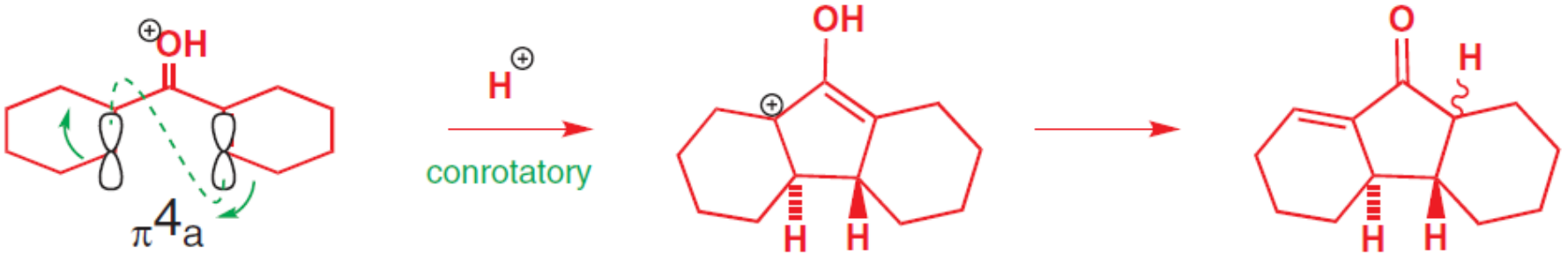


Nazarov cyclizations: stereochemistry

- Thermal conditions

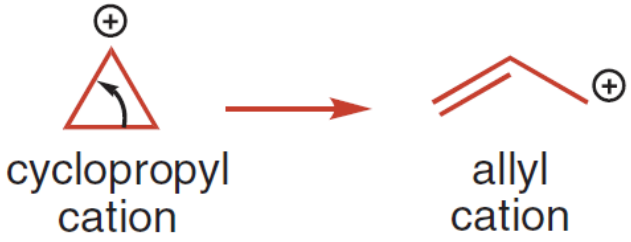
Is the reaction disrotatory or conrotatory?

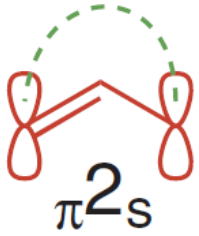




6. Small ring opening cyclopropyl cation

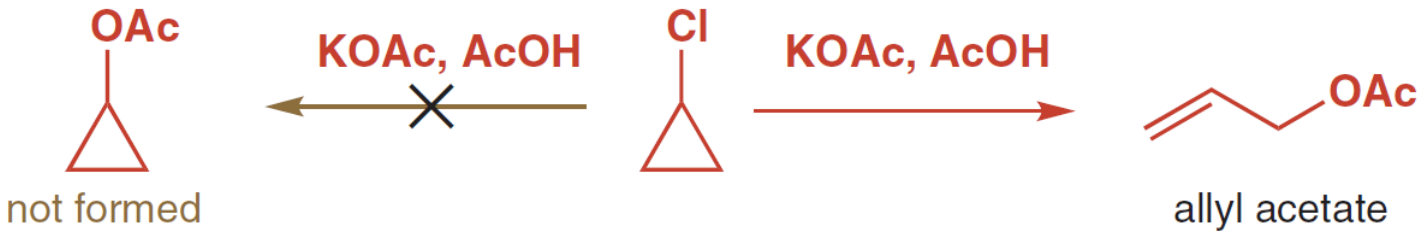
Is the ring opening of cyclopropyl cation disrotatory or conrotatory?





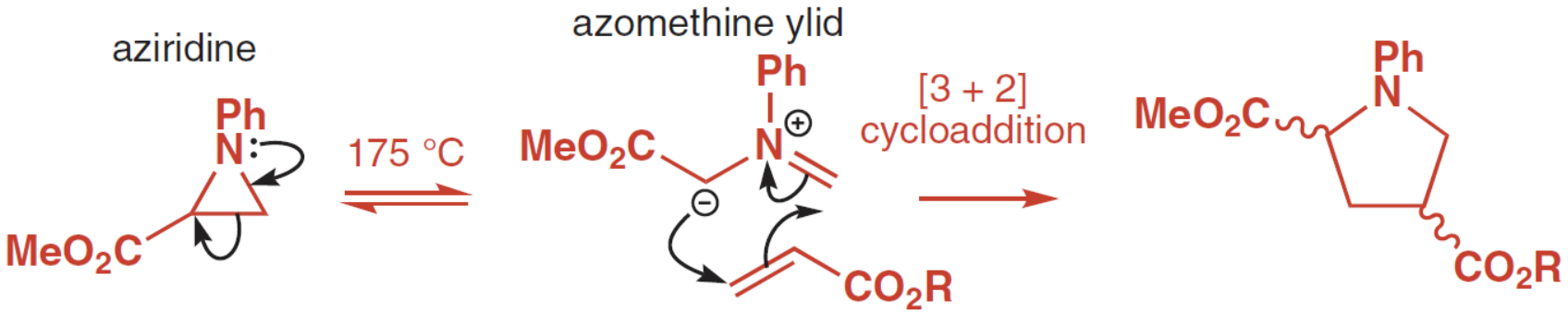
2e, disrotatory

Instability of cyclopropyl cations: ring strain as a driving force



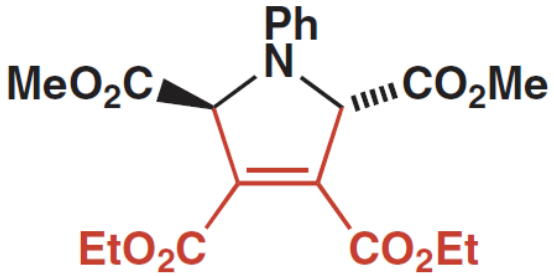
Small ring opening aziridine

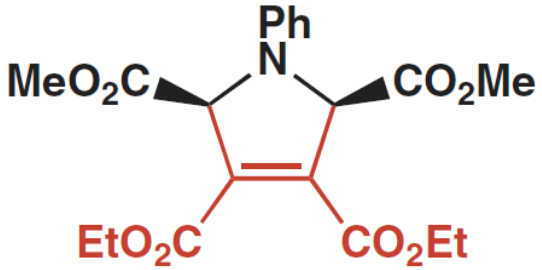
Is the ring opening of aziridine disrotatory or conrotatory?



4e, conrotatory

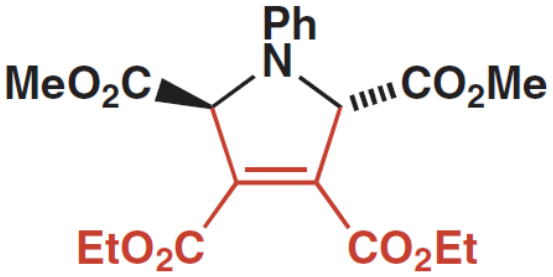
Proposed the stereochemistry of the starting aziridines for the preparation of the following products.

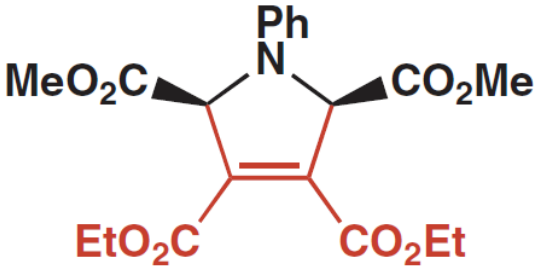




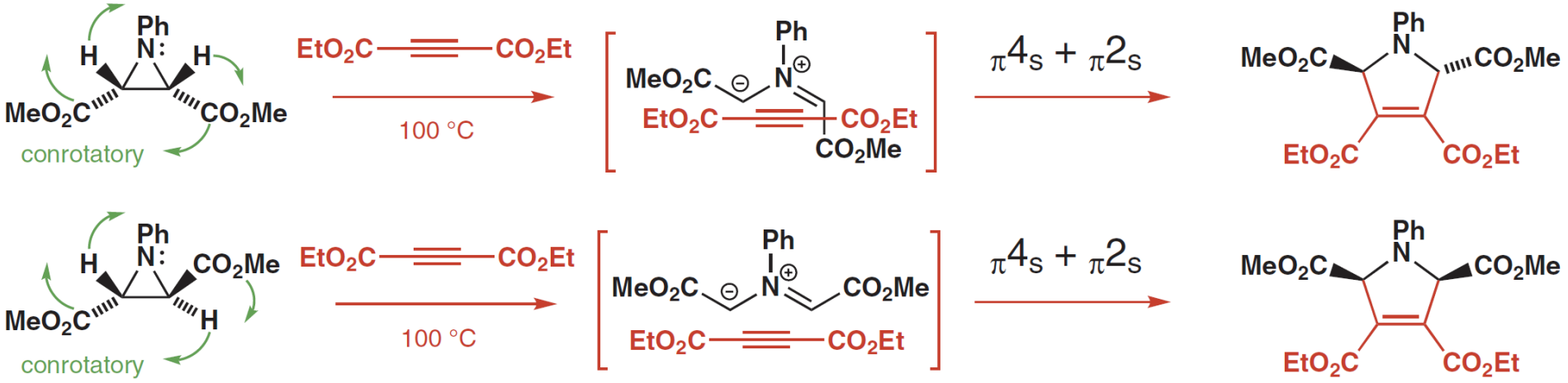
Small ring opening aziridine

Proposed the stereochemistry of the starting aziridines for the preparation of the following products.





4 pi electrons under thermal conditions→ conrotatory.



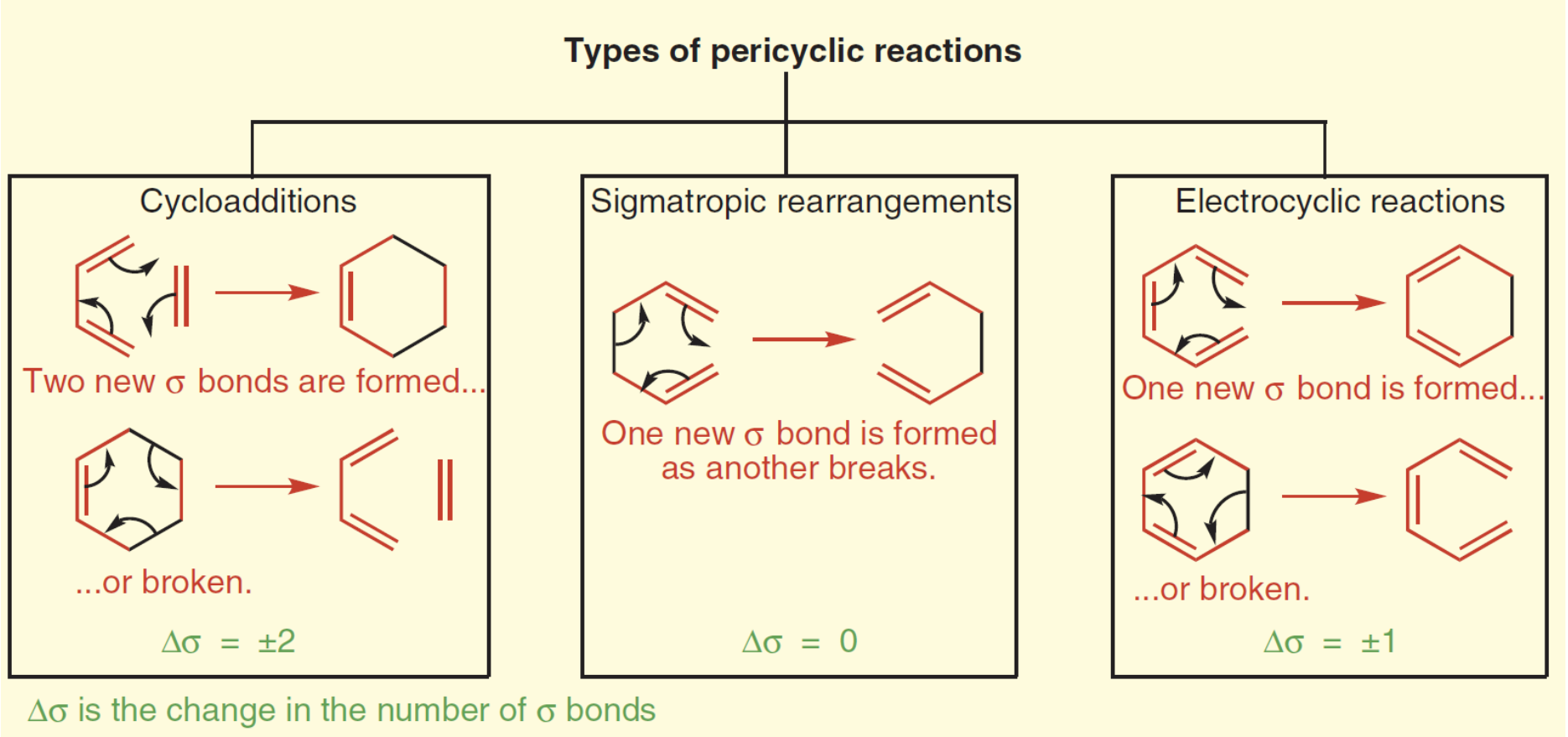
Summary: electrocyclic reactions

- Applying Woodward–Hoffman rules
- Disrotatory and Conrotatory
- Thermal and photochemical
  - 8π tetraene
  - 6π triene
  - 4π diene, Nazarov
  - Small ring opening (2π cyclopropyl cation, 4π aziridine)

Pi System	Thermal	Photochemical
"4n" electrons	Conrotatory	Disrotatory
"4n + 2" electrons	Disrotatory	Conrotatory

Disrotatory : **suprafacial** Conrotatory: **antarafacial**

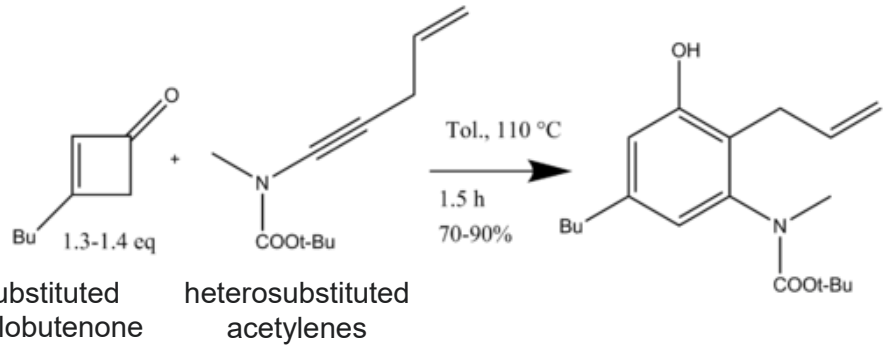
Pericyclic reactions: Change in the number of sigma bonds



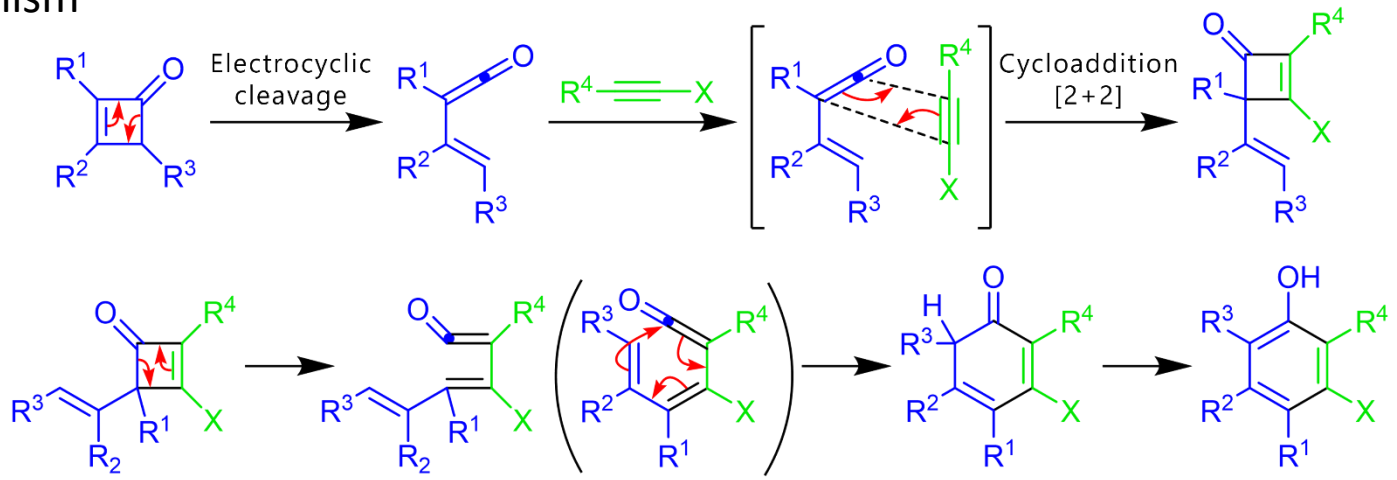
Both forward and reverse reactions follow Woodward–Hoffman rules.

Danheiser benzannulation

Application: synthesis of poly-substituted benzenes



mechanism



The synthesis of a cockroach pheromone

